

Evaluating Open-Weight LLMs for Turkish Domain Documents Under Retrieval and Hardware Constraints

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Abstract

Most Turkish-capable large language models (LLMs) are evaluated using general-purpose benchmarks rather than long, structurally complex domain documents. This paper evaluates five open-weight 7B–8B models for Turkish document question answering under a resource-constrained local deployment setting. The primary benchmark contains 100 systematically validated questions derived from a 109-page industrial R&D report, and the evaluation protocol is replicated using a second 112-page public-sector report and an independently constructed 100-question set. All models are evaluated locally on an NVIDIA RTX 3050 laptop GPU with 6 GB VRAM using controlled prompting, decoding, and 4-bit quantisation.

The principal methodological contribution is an evidence-annotated evaluation protocol that separates retrieval failure from downstream model reasoning failure without requiring additional model calls. On the primary benchmark, end-to-end accuracy ranges from 49% to 75%. Seven lexical, dense, and hybrid retrieval configurations are additionally compared using 95% Wilson intervals and exact paired McNemar tests; none significantly outperforms the character TF-IDF baseline on either document. Evidence recall saturates differently across the two reports, showing that retrieval and effective context capacity can be binding constraints for some documents but not others. These results demonstrate that model selection, retrieval behaviour, and hardware limits must be evaluated separately when deploying open-weight LLMs for Turkish domain documents.

Index Terms

Large language models, Turkish NLP, document-grounded question answering, retrieval-augmented generation, model evaluation, local deployment

I. INTRODUCTION

The set of Turkish-capable open-weight LLMs has grown rapidly, with Turkish-tuned models appearing alongside increasingly capable multilingual systems [6]–[8], [13]. However, their effectiveness on realistic Turkish document-analysis tasks remains unclear. Existing evaluations rely largely on educational, general-knowledge, multiple-choice, and curated reading-comprehension tasks [1]–[5]. These settings generally do not evaluate long, structurally heterogeneous Turkish documents processed through a retrieval pipeline, which is a common deployment scenario for document-grounded LLM systems. We therefore ask whether current open-weight LLMs can reliably support Turkish document-grounded question answering under realistic local deployment constraints.

This question is particularly important for confidential industrial documents, where cloud processing may conflict with organisational or data-governance requirements. Such documents may originate in heterogeneous formats but are commonly distributed as PDF reports containing text, tables, and structural layout that must be recovered before retrieval. This setting is relevant to the SynAM project, which combines data, machine learning, and engineering research across academic and industrial partners. We therefore study a Turkish industrial PDF using a locally deployed pipeline on a consumer GPU.

Answering this question requires separating two failure modes that conventional end-to-end evaluations often conflate. When a model answers a document question incorrectly, either the model failed to reason over the supplied evidence, or the retrieval system failed to provide the necessary evidence. Treating pipeline accuracy as model accuracy silently charges the model for the retriever’s misses, even though the two failure modes require different remedies. Recording, for each question, the specific evidence on which the answer depends allows evidence recall to be measured independently of model output and enables errors to be attributed either to the model or to the retrieval apparatus. This decomposition becomes central to the analysis presented in this paper.

We make three contributions. First, we construct and validate two document-specific Turkish QA benchmarks, each containing 100 questions covering factual retrieval, numerical reasoning, multi-step inference, and abstention when the requested information is absent. The first benchmark is based on a 109-page industrial R&D report, while the second is based on a 112-page public-sector activity report. Each question records its supporting evidence, enabling retrieval failures to be distinguished from model failures.

Second, we provide a controlled comparison of five open-weight 7B–8B models under identical retrieval, prompting, decoding, quantisation, and hardware conditions, using a question-specific grading procedure that accounts for equivalent answer forms.

Third, we compare character and word TF-IDF, BM25, dense retrieval, and hybrid retrieval under the same chunking and context-budget conditions. Evidence recall is reported with 95% Wilson intervals, and paired McNemar tests are used to compare the retrieval methods. The results show that none of the evaluated alternatives significantly outperforms character TF-IDF on either document. They also show that the effect of limited context capacity differs between the two documents.

II. RELATED WORK

Turkish LLM evaluation. Existing Turkish LLM evaluations focus primarily on general knowledge, educational content, and multiple-choice reasoning. MMLU [15] established the widely used multitask multiple-choice evaluation paradigm. Turkish-MMLU [2] applies this paradigm using more than 10,000 native Turkish questions drawn from the secondary-school curriculum, while TR-MMLU [1] evaluates models using 6,200 Turkish multiple-choice questions spanning educational and domain-specific categories. The OpenLLM Turkish Leaderboard [3] aggregates evaluations based largely on established general-purpose tasks, and Dogan et al. [4] compare Turkish-capable models across several such benchmarks. THQuAD [5] moves beyond multiple-choice evaluation by testing reading comprehension over curated Turkish historical passages. These resources provide valuable evidence about Turkish language understanding and general knowledge, but they do not evaluate question answering over a long, structurally heterogeneous domain document processed through a retrieval pipeline.

Document-grounded QA and retrieval. Retrieval-augmented generation [14] is a widely used architecture for grounding language-model outputs in external documents. In such systems, end-to-end accuracy depends not only on the generator but also on whether the retriever supplies the evidence required to answer the question. Reporting only final answer accuracy can therefore conflate retrieval failure with generation or reasoning failure. The present work addresses this issue by recording the supporting evidence required for each question and measuring whether that evidence appears in the retrieved context.

This distinction is especially relevant under a fixed context window. Turkish is morphologically rich, and its word forms may be segmented into relatively large numbers of tokens by tokenisers developed primarily for multilingual or English-dominant training corpora. In our deployment setting, this reduces the amount of document text that can be supplied within the available context budget. We therefore examine retrieval quality and effective context capacity as components of the evaluation apparatus rather than treating them as implementation details.

Grading of generative QA. Open-ended QA cannot always be evaluated reliably by exact matching because equivalent answers may differ in wording, inflection, numeric formatting, or date representation. We therefore use question-specific grading rules for numeric, entity, keyword, date, Boolean, multiple-choice, and abstention answers, as detailed in Sec. III-E.

These limitations motivate an end-to-end evaluation beginning with the original PDF and including extraction, retrieval, generation, and grading.

III. METHODOLOGY

A. Benchmark Design

The main evaluation uses a 109-page Turkish industrial R&D report in PDF format containing running text, personnel and financial tables, publication records, patents, collaborations, and project descriptions. The PDF is converted to layout-preserving text before sectioning, chunking, and retrieval. From this report, we construct and verify a 100-question benchmark.

To examine whether the evaluation procedure and the main observations depend entirely on one document, we conduct a secondary cross-document validation using the 112-page 2024 activity report of the Turkish Personal Data Protection Authority [16]. This public-sector report differs from the primary document in its source, structure, and subject matter. A separate 100-question set is constructed using the same question structure and evaluation procedure. This second evaluation is used as a robustness check rather than as a comprehensive multi-domain benchmark.

For each document, the questions are divided into four capability bands comprising 30 Easy factual-retrieval questions, 30 Medium numerical-reasoning questions, 20 Hard multi-step questions, and 20 Abstention questions whose answers are not present in the corresponding report. Each question records its expected answer, supporting evidence, and grading rule. The first three bands test increasingly demanding use of retrieved evidence, while the Abstention band tests whether a model avoids producing unsupported answers.

Figure 1 summarises the complete evaluation pipeline from document preprocessing and question-aware retrieval to prompt construction, model inference, evidence-recall analysis, and answer grading.

B. Evaluation Protocol

All models are evaluated under identical conditions. For a given question, the retrieval stage is executed once and the resulting context is kept fixed across all five models. Consequently, each model receives the same question, retrieved evidence,

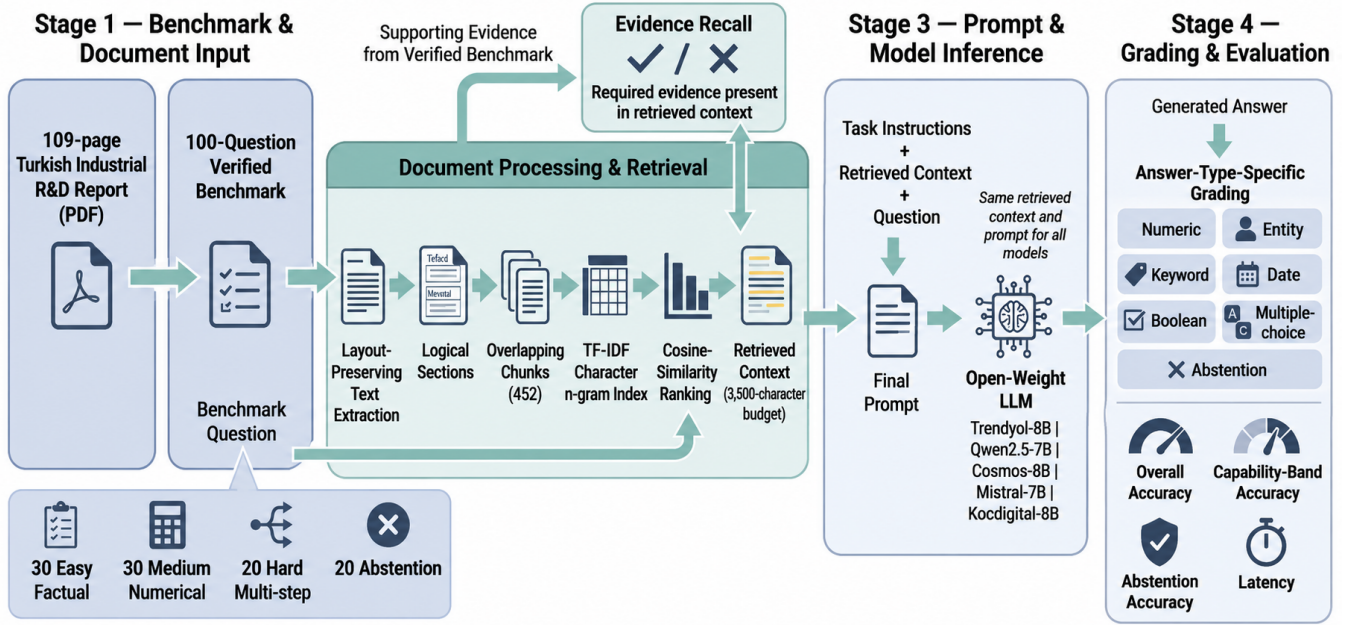


Fig. 1. End-to-end evaluation pipeline. The same retrieved context and prompt are supplied to all models, while evidence recall is evaluated independently from answer grading.

prompt structure, decoding settings, and context budget. This allows differences in answer quality to be compared without introducing model-specific differences in retrieval.

Inference is performed locally on an NVIDIA RTX 3050 laptop GPU with 6 GB VRAM. The 4,096-token context window was selected as a stable common setting that fit within the available memory budget after accounting for quantised model weights, KV-cache memory, prompt tokens, and generated output. All models use temperature 0 and a fixed seed and are served locally via Ollama.

We report overall accuracy, accuracy by capability band, abstention accuracy, evidence recall, and latency. Overall accuracy is the proportion of the 100 questions answered correctly, while capability-band accuracy is calculated within each question subset. Evidence recall is measured separately from answer accuracy and indicates whether the evidence required to answer a question is present in the retrieved context. This distinction allows retrieval failures to be separated from failures that occur after the relevant evidence has been supplied to the model. The primary evaluation and the secondary cross-document validation use the same model configurations, question distribution, prompting procedure, decoding settings, and context budget. Unless otherwise stated, the model comparisons use the character TF-IDF retriever described in Sec. III-D. The retrieval comparison changes only the ranking method while keeping chunking, context selection, and the character budget fixed.

Evidence recall is calculated only for questions whose required supporting figures are explicitly recorded and can be checked in the retrieved context. Abstention questions are excluded because the requested information is absent by construction. Questions whose expected answer is a Boolean value or a multiple-choice label are also excluded when that label does not represent the supporting evidence in the document.

Uncertainty in evidence-recall estimates is reported using 95% Wilson score intervals. Retrieval methods are compared using exact paired McNemar tests because each method is evaluated on the same questions. A difference is treated as statistically significant when $p < 0.05$.

C. Evaluated Models

We evaluate five open-weight models in the 7B–8B parameter range: Trendyol-LLM-8B-T1, Qwen2.5-7B, Cosmos Turkish-Llama-8B-Instruct, Mistral-7B, and Kocdigital-LLM-8B. Trendyol-LLM-8B-T1 is a Turkish–English reasoning model built on Qwen3-8B [6]. Qwen2.5-7B and Mistral-7B serve as general multilingual baselines [8], [11]. Cosmos and Kocdigital are Turkish-adapted Llama-based models [9], [10], [12].

The models were selected because they have comparable parameter counts, represent both Turkish-specialised and general multilingual systems, and can be evaluated using the same Q4_K_M quantisation and 6 GB GPU constraint. Their parameter counts are reported in Table I, while the cited model cards provide the corresponding model configurations.

D. Document Processing, Retrieval, and Prompt Construction

As shown in Fig. 1, each source PDF is converted to text using layout-preserving extraction so that line structure and the relative arrangement of table content are retained as far as possible. The extracted text is organised into logical sections and divided into overlapping chunks. This produces 452 chunks for the primary industrial report and 348 chunks for the secondary public-sector report.

The main model evaluation uses TF-IDF character n -gram features. For each benchmark question, the question text is represented in the same feature space as the document chunks. Cosine similarity is calculated between the question and each chunk, producing a ranked list of candidate evidence. Chunks are selected in descending similarity order until the 3,500-character context budget is reached.

For the retrieval comparison, seven configurations are evaluated under the same chunking and context-selection conditions. These comprise character TF-IDF, word TF-IDF, BM25, multilingual E5-small, multilingual E5-base, reciprocal-rank fusion of character TF-IDF and BM25, and reciprocal-rank fusion of character TF-IDF and E5-base. Only the ranking method is changed. The chunk boundaries, context budget, overlap handling, and document-order assembly remain fixed.

The selected chunks form the evidence context for the question. The final prompt consists of task instructions, the retrieved context, and the question itself. The instructions require the model to answer using the supplied document evidence and to abstain when the requested information cannot be supported. For a given document and question, the same assembled prompt is supplied to each of the five models.

E. Grading and Score Calculation

Exact string matching is unsuitable for model-generated answers because semantically equivalent Turkish responses may differ in number formatting, date representation, inflection, wording, or explanatory text. Each benchmark question is therefore assigned an answer-type-specific grading rule before model evaluation. The supported answer types include numeric, multiple-choice, Boolean, entity, keyword, date, and abstention responses.

The grading rule defines what constitutes a correct response for each question. Numeric questions are judged using the required numerical value rather than the surrounding explanation. Years and exact integer counts require an exact value, while a small numerical tolerance is retained for derived ratios, percentages, and rounded decimal values. Entity and keyword questions are checked against predefined acceptable forms. Multiple-choice and Boolean questions are mapped to their expected category, and Abstention questions are scored as correct only when the model indicates that the requested information cannot be established from the supplied evidence. Each question receives a binary score of 1 for a correct answer and 0 otherwise.

Overall accuracy is the proportion of correctly answered questions, with the same calculation applied within each capability band. Evidence recall is computed independently from the expected answer string. A question is counted as recalled only when all recorded source figures required to produce its answer are present in the retrieved context. This is particularly important for numerical-reasoning questions whose final answers are calculated and may not appear literally in the source document.

The reference answers and supporting evidence were checked against the cited pages of the corresponding source document. Derived answers were recalculated from the source tables, while the targets of Abstention questions were checked using full-document search. The configured grading rules were also tested using the expected answers before model outputs were evaluated.

IV. RESULTS

A. Accuracy and Inference Cost

Fig. 2 and Table I show that overall accuracy ranges from 49% to 75%, a 26-point spread despite comparable model sizes and identical evaluation conditions. Table II reports latency measured on the fixed 20-question timing sample.

TABLE I
BENCHMARK ACCURACY BY CAPABILITY BAND.

Model	Par.	Acc.	E /30	M /30	H /20	Abs /20
Trendyol-8B	8.2B	75%	21	22	14	18
Qwen2.5-7B	7.6B	65%	21	15	10	19
Cosmos-8B	8.0B	54%	12	12	12	18
Mistral-7B	7.2B	51%	16	12	10	13
Kocdigital-8B	8.0B	49%	19	12	9	9

TABLE II
INFERENCE LATENCY ON THE TIMING SAMPLE.

Model	Mean (s/q)	Median (s/q)	Est. 100 q (min)	s/correct
Trendyol-8B	23.86	23.36	39.8	31.8
Qwen2.5-7B	1.69	1.47	2.8	2.6
Cosmos-8B	3.21	3.21	5.4	6.0
Mistral-7B	2.63	2.42	4.4	5.2
Kocdigital-8B	5.75	3.30	9.6	11.7

Trendyol and Qwen2.5 tie on Easy questions at 21/30, but Trendyol reaches 22/30 on Medium questions compared with 15/30 for Qwen2.5 and 12/30 for the remaining models. This pattern suggests that reasoning over retrieved figures, rather than

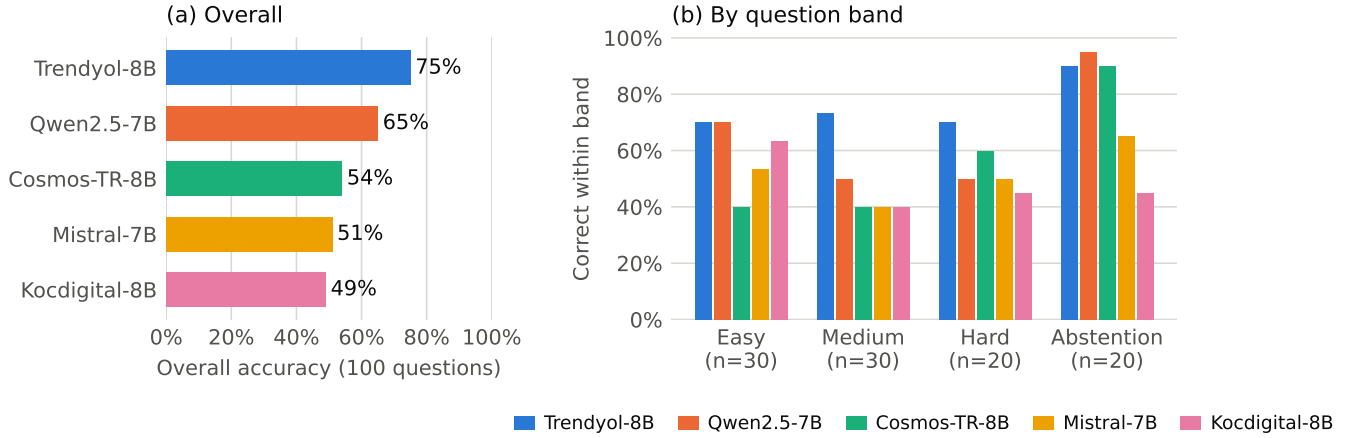


Fig. 2. Model performance. (a) Overall accuracy on the 100-question benchmark. (b) Accuracy by capability band. Trendyol leads overall, with its largest advantage in the Medium band, while abstention performance reveals a different ordering from factual accuracy.

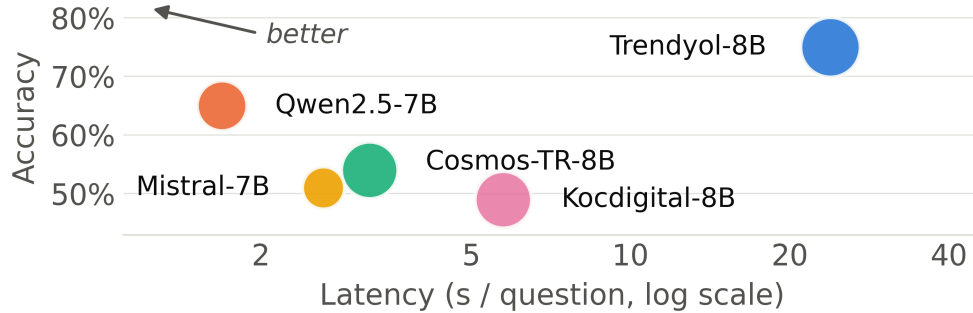


Fig. 3. Accuracy–latency trade-off on the target 6 GB GPU. The upper-left region is preferable.

extraction alone, contributes to Trendyol’s advantage. The multilingual Qwen2.5 model outperforms Cosmos and Kocdigital, showing that Turkish-specific adaptation alone does not guarantee stronger document-grounded performance.

The secondary cross-document evaluation produced overall accuracies of 82%, 65%, 60%, 58%, and 51% for Trendyol, Qwen2.5, Cosmos, Kocdigital, and Mistral, respectively. The three strongest models retained their ordering from the primary benchmark, while the two lowest-ranked models exchanged positions.

B. Faithfulness

Abstention performance differs from overall accuracy. Qwen2.5 achieves the highest abstention score at 19/20, while Cosmos and Trendyol each reach 18/20. Kocdigital correctly abstains on only 9/20 questions, despite scoring 19/30 on Easy questions. This makes it less suitable for factual report-search settings where unsupported answers are costly.

Qwen2.5 provides the strongest efficiency trade-off, achieving the highest accuracy with the lowest mean latency. Trendyol gains ten accuracy points but is approximately fourteen times slower per question. Its final answers contain 207 characters on average after scratchpad removal, compared with 44 for Qwen2.5. Trendyol-T1 also inherits Qwen3’s thinking mode, which generates intermediate reasoning tokens before the final answer. These tokens are removed before grading and output-length measurement but still contribute to inference latency. Extraction, indexing, retrieval, and grading contribute negligible runtime compared with model inference.

C. Retrieval Comparison

Seven retrieval configurations were compared using the same chunking and 3,500-character context budget. On the primary report, evidence recall was 57% for character TF-IDF, 65% for BM25, 59% for E5-base, and 63% for the strongest hybrid. On the secondary report, the corresponding values were 91%, 86%, 90%, and 91%. The character TF-IDF estimates had 95% Wilson intervals of [44, 70] and [82, 96], respectively. Exact paired McNemar tests found no significant improvement from any lexical, dense, or hybrid alternative on either document, with all $p > 0.05$. The results therefore show no consistent advantage for a more complex retriever under the evaluated conditions.

V. ERROR DECOMPOSITION

End-to-end accuracy includes both model and retrieval failures. We separate these cases by checking whether all source figures required for an answer are present in the retrieved context.

At the 3,500-character operating budget, evidence recall is 31/54, or 57%, on the primary report and 70/77, or 91%, on the secondary report. Primary recall is 74% for Easy, 43% for Medium, and 58% for Hard questions. The corresponding secondary results are 96%, 93%, and 79%. Only questions with explicitly recorded source figures are included. Abstention questions and Boolean or multiple-choice labels without retrievable source figures are excluded. Differences in evidence annotation mean that the two overall values should not be interpreted as a controlled ranking of the documents.

Diagnostic budgets of 7,000 and 9,000 characters increase primary recall to 72% and 76%, while secondary recall changes only to 94% and 95%. These larger budgets contain the evidence selected at smaller budgets and therefore indicate where recall saturates. They are not used for model evaluation because a 7,000-character context plus the generation allowance exceeds the common 4,096-token limit. Retrieval is therefore a substantial constraint for the primary report but is less restrictive for the secondary report.

VI. CONCLUSION

We evaluated five open-weight 7B–8B models for Turkish document QA under a fixed retrieval pipeline and a 6 GB consumer-GPU constraint. On the primary industrial report, accuracy ranged from 49% to 75%, with Trendyol strongest overall and Qwen2.5 providing the best accuracy–latency trade-off. On the secondary public-sector report, the corresponding model accuracies ranged from 51% to 82%, and the three strongest models retained their ordering.

The evidence-based analysis showed 57% retrieval recall on the primary report and 91% on the secondary report at the same operating budget. None of the evaluated lexical, dense, or hybrid retrieval alternatives produced a significant improvement over character TF-IDF. These results show that the effect of retrieval and context limitations depends on the source document.

The primary benchmark remains based on one industrial R&D report, while the public-sector report serves as a cross-document robustness check. Replication using legal, medical, and other sector-specific documents is needed before broader conclusions can be drawn.

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